

PROFESSIONAL DEVELOPMENT ROADMAP

Academic Year 2026–27

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1. JOB DESCRIPTION

AICTE IDEA Lab Responsibilities

- Planning and execution of IDEA Lab activities.
- Equipment utilization and maintenance.
- Innovation ecosystem development.
- Product development support.
- Student project mentoring.
- Procurement and inventory management.
- Utilization reporting and documentation.

Target

- Events to be Conducted: 10
- Students Trained: 500+
- Product Development Projects: 15
- Equipment Utilization: >80%

R Fab X Studio Responsibilities

- Fabrication support.
- Rapid prototyping.
- Industry consultancy.
- Student training.
- Startup incubation support.
- Technology demonstrations.

Target

- Technical Workshops: 6
- Consultancy Projects: 1
- Prototype Development Projects: 4
- Industry Training Programs: 2

2. NPTEL, FDP & CERTIFICATION TARGETS

NPTEL Courses

Target: 3 Courses

Suggested Courses: 1. Product Design and Development 2. Additive Manufacturing 3. Innovation and Entrepreneurship

Expected Outcome

- Three Elite Certification

Faculty Development Programmes

Target: 5 FDPs

Areas:

- AI Tools in Education
- Robotics
- Industry 4.0
- Additive Manufacturing
- Product Design

3. RESEARCH DEVELOPMENT PLAN

Journal Publications

Target

- Scopus Indexed Papers: 1
- Peer Reviewed Papers: 1

Proposed Research Areas

- Metal Matrix Composites
- Additive Manufacturing
- Design Optimization
- Sustainable Manufacturing

Conference Publications

Target

- International Conferences: 2

Research Proposal Submission

Target

- Funding Proposal: 1

Possible Agencies:

- AICTE
- ATAL
- DST
- MSME
- TNSCST

4. INTELLECTUAL PROPERTY RIGHTS

Patent Targets

Patent Filing

Target: 3

Patent Publication

Target: 2

Patent Grant

Target: 1

Copyright Filing

Target: 2

Possible Areas:

- Educational Innovation
- Manufacturing Systems
- Robotics Applications
- Student Training Kits

5. AICTE IDEA LAB ACTION PLAN

Workshops

Target: 10

Possible Topics:

- 3D Printing
- CAD Modelling
- Robotics
- Arduino Applications
- CNC Programming
- Product Design

Innovation Activities

Activity	Target
Innovation Challenges	2
Design Thinking Workshops	2
Product Development Projects	15
Faculty Innovation Programs	2
Open House Events	1

Outreach Programs

Target: 3

Participants:

- School Students
- Polytechnic Students
- Engineering Students

6. R FAB X STUDIO ACTION PLAN

Technical Workshops

Target: 6

Topics:

- Welding
- Carpentry
- Sheet Metal
- Robotics
- CNC Machining
- Product Development

Consultancy Services

Target: 1 Project

Areas:

- CAD Design
- Product Development
- Reverse Engineering

Revenue Generation

Consultancy Revenue

₹ 50,0000

Training Revenue

₹4,00,000

Prototype Revenue

₹50,000

Total Target

₹5,00,000

7. STARTUP DEVELOPMENT

Startup Support

Target: 2 Teams

Activities:

- Product Validation
- Prototype Development
- Business Model Development

Expected Outcome

- Startup Registrations: 1
- Commercialized Products: 1

8. STUDENT INNOVATION & COMPETITION MENTORING

Smart India Hackathon

Activity	Target
Teams Mentored	5
Internal Hackathon	1
Teams Shortlisted	2

YUDHISTRA

Activity	Target
Teams Mentored	10
Projects Developed	5

Design Contests

Activity	Target
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Activity	Target
Teams Mentored	3
Prototypes Developed	3

Additional Innovation Activities

Activity	Target
Innovation Teams Formed	10
Prototypes Developed	15
Student Innovation Projects	20
Competition Participation	15 Teams

9. ARIVUPATTARAI 2026–27

Departments Covered

Target: 4

Student Beneficiaries

2200

Student Volunteers

50

Expected Outcome

- Enhanced hands-on learning.
- Improved interdisciplinary collaboration.
- Product development culture.

10. INDUSTRY COLLABORATION

Activity	Target
Industrial Visits	2
Industry Experts Invited	6
Industry Training Programs	2
Industry Consultancy Projects	3

11. DIGITAL CONTENT CREATION

YouTube Channel Development

Target:

- Educational Videos: 25
- Lab Demonstration Videos: 10
- Innovation Showcase Videos: 5

12. COMMUNITY & SOCIAL IMPACT

Activities:

- School Outreach Programs
- STEM Awareness Programs
- Rural Innovation Awareness
- Entrepreneurship Awareness

Target:

- Beneficiaries: 300+

13. KEY PERFORMANCE INDICATORS (KPIs)

KPI	Target
Scopus Papers	3
Patents Filed	3
Patent Publications	2
NPTEL Courses	3
FDPs	5
Workshops	16
Students Trained	750+
Innovation Projects	20
Startup Teams	5
Revenue Generated	₹5,00,000

KPI	Target
SIH Teams	5
Yudhistra Teams	5
Design Contest Teams	3
Outreach Programs	3
Industrial Collaborations	2

14. EXPECTED ACHIEVEMENTS BY MAY 2027

- Successful execution of Arivupattarai 2026–27.
- AICTE IDEA Lab recognized as a high-utilization facility.
- Filing of 3 patents and publication of 2 patents.
- Publication of 1 Scopus-indexed research papers.
- Completion of 3 NPTEL certifications.
- Generation of ₹5 Lakhs through consultancy and training.
- Mentoring of 5 startup teams.
- Training of more than 2200 students in the Arivupattarai.
- Development of multiple prototypes and innovative products.
- Strengthening RTC's reputation in innovation, entrepreneurship, and experiential learning.